

The Beacon Theorem

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The Beacon Terminology

The beacon is a powerful block used to provide enchantments to the player. The beacon is a crafted block requiring items to obtain; however, it is not easy for a player to have the needed requirements to craft a beacon. It requires a nether star which is obtained by the murder of the wither boss that is the second powerful mob after the warden on Java Edition and the most powerful mob on Bedrock Edition.

To craft a beacon, it is required to have 5 glasses, 3 obsidian blocks, and a nether star. First, the 3 glasses should be placed in the top row of the crafting table, the remaining glasses should be placed in the middle row in the left and the right column. Then, the 3 obsidian blocks should be placed in the bottom row. Finally, a nether star should be placed in the middle column and row.

The Blocks Required for a Beacon Layer

The beacon pyramid consists of blocks with an area. Each layer is multiplied. This means that the layers below are getting multiplied. For example, a beacon with 3 layers will require less blocks in the 1st layer, more blocks in the 2nd layer than the 1st layer, and more blocks in the last layer than the 2nd layer. The last layer shares the largest area of the layers. It is just like a staired pyramid. The more the layer increases, the more the choices of the effects increase, and the higher the range of the effects becomes.

There are certain blocks required to build a beacon. Those blocks activate the beacon when it is built correctly. The beacon should be placed correctly, and the materials should be within the required blocks. The beacon requires iron blocks, gold blocks, diamond blocks, emerald blocks, and netherite blocks in order to active.

Figure 1 shows the possible blocks for a beacon to activate. This shows how effortful it is to craft a beacon, and to have adequate blocks for a layer.



Figure 1: The materials needed for a beacon to activate.

Minerals Required for a Beacon

The beacon activates by the required blocks needed; however, there are minerals required to put inside to activate the effects. Those minerals are iron ingots, gold ingots, diamonds, emeralds, and netherite ingots. They can be one of those items to activate an effect.

Figure 2 shows what is inside the beacon and the required items to place.



Figure 2: The required minerals to activate an effect.

The First Beacon Formula

There are procedures that should be followed to build a beacon. The first beacon formula is the formula of the beacon if the maximum beacons added was only one. This means that the layers should be a perfect odd square to activate the beacon. If the square numbers are 1, 4, 9, 16, 25, 36, 49, etc..., this means that it should be 1, 9, 25, 49, etcetera.

$$b_l = h^2$$

This makes b_l equals a square number where it can be even or odd. For the formula to be valid, b_l has to be an odd number where $h \in \mathbb{N}_0$.

$$b_l = (2h)^2$$

This makes b_l a square even number where $h \in \mathbb{N}_0$.

$$b_l = (2h + 1)^2$$

Finally, this makes b_l a square odd number where $h \in \mathbb{N}_0$.

The First Beacon Layer Formula

$$b_l = (2h + 1)^2$$
$$h \in \mathbb{N}_0$$

Where h is the number of the layer in the beacon and b_l is the number of the blocks in layer h which is the area of layer h . When $h = 0$, b_l becomes the maximum numbers of beacons that can be added measured in blocks² or the area of the beacons. Since this is the first beacon formula, $b_l = 1$ blocks² indicating that only one beacon is applicable to be placed in the pyramid to activate it.

The beacon layer formula only calculates the total blocks in a specific layer. To calculate the total blocks needed in the entire pyramid

including the beacon, take the summation of the layer formula with intervals.

$$b = \sum_{h=0}^m b_l$$

Substituting $b_l = (2h + 1)^2$ generates the full formula of the beacon.

The First Beacon Formula

$$b = \sum_{h=0}^m (2h + 1)^2$$

$$h \in \mathbb{N}_0$$

$$m \in \mathbb{N}$$

b is the total numbers of blocks needed to build the entire beacon measured in blocks² while m is the maximum layer or the final layer that is decided by the player. The players can choose the layer from layer 1 to their desired layer as long as it does not reach the maximum building limit.

To find the blocks needed without the beacon itself, subtract the summation by the number of beacons which is 1 in this case. This helps in organizing the minerals needed to craft the blocks as an initiation of the pyramid layers, then finalize with the beacon. **Figure 3** shows the division of the beacon layers.

The Sum of Layers Excluding the Beacon

$$\sigma_l = b - 1$$

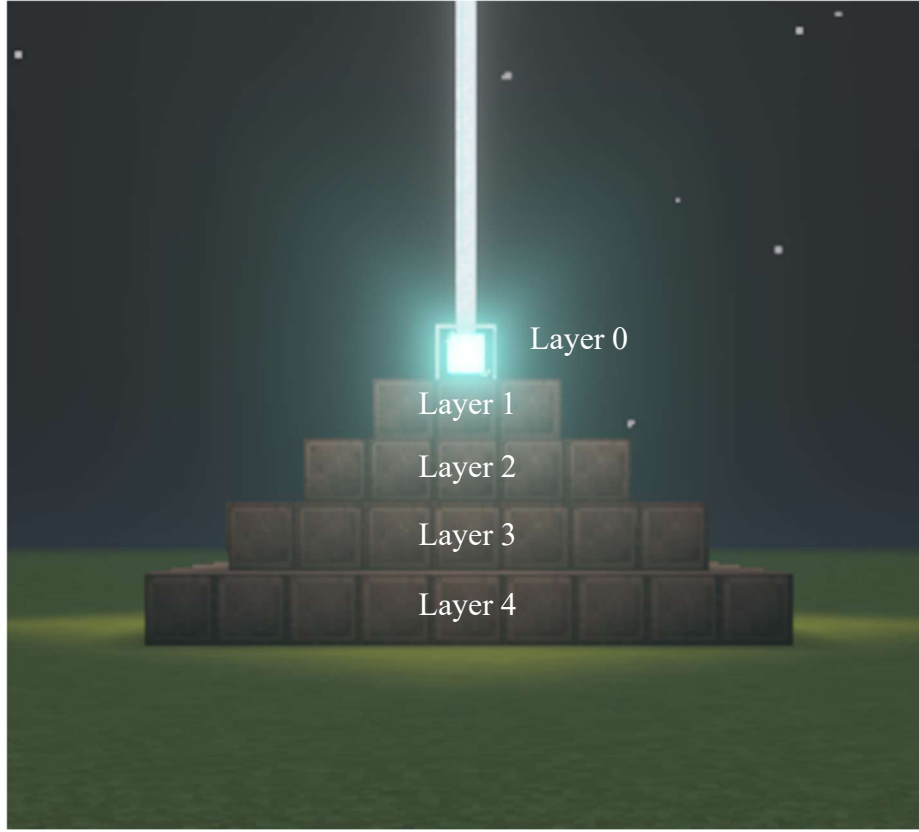


Figure 3: The division of the beacon layers.

The Second Beacon Formula

The second beacon formula is the formula of the beacon if the maximum number of beacons is a perfect square area. The maximum number of beacons can be more than just a beacon. It can be 4, 9, 16, 25, 36, 49, etc... blocks². Take the first formula, then adjust it.

$$b_l = (2h + 1)^2$$

The 1 in the formula indicates the side of the maximum beacon since the equation is a perfect square. Let 1 becomes a variable where the maximum number of beacons in a side can be any $\mathbb{N}$.

$$b_l = (2h + s)^2$$

This makes s the number of the maximum beacons in a side measured in blocks.

The Second Beacon Layer Formula

$$b_l = (2h + s)^2$$

$$h \in \mathbb{N}_0$$

$$s \in \mathbb{N}$$

s is the number of the maximum beacons in a side. To find the area of the maximum beacons or the total maximum beacons, this equation must be utilized.

$$A_s = s^2$$

Where A_s is the total maximum beacons measured in blocks². If h is 0, the $b_l = A_s$ making it equals the total maximum beacons.

To find the number of the blocks in a beacon layer's side, square root the b_l .

$$S_B = \sqrt{b_l}$$

Plug in the function of b_l to generate the side formula.

The Beacon Layer Side Formula

$$S_B = 2h + s$$

$$h \in \mathbb{N}_0$$

$$s \in \mathbb{N}$$

Where S_B is the side of the beacon layer measured in blocks. If $h = 0$, $S_B = s$ making it equals the total maximum beacons in a side.

There is a formula that also calculates the sum of the layers and the beacons using the same formula; however, s will be added instead of 1.

$$b = \sum_{h=0}^m b_l$$

Let $b_l = (2h + s)^2$.

$$b = \sum_{h=0}^m (2h + s)^2$$

This only calculates the total layers and the total maximum beacons; however, the number of beacons can be less than the maximum. This changes the equation adding a new variable to the summation formula.

The Second Beacon Formula

$$b = \sum_{h=0}^m (2h + s)^2 - \beta$$

$$h \in \mathbb{N}_0$$

$$s \in \mathbb{N}$$

$$m \in \mathbb{N}$$

$$\beta \in \mathbb{N}_0 \mid 0 \leq \beta < A_s$$

In the second beacon formula, β is the number of beacons deducted which is decided by the player. This makes the second formula

different from the first one because it was assumed in the first formula that $A_s = 1 \text{ blocks}^2$, this automatically makes $\beta = 0$. For the second formula, $A_s \geq 1 \text{ blocks}^2$ which turns β into a variable.

To calculate the beacons deducted, subtract the total maximum beacons by the remaining beacons.

The Equation of the Deducted Beacons

$$\beta = A_s - n_b$$

$$n_b \in \mathbb{N} \mid n_b \leq A_s$$

n_b is the number of the beacons decided by the players. If there are no deducted beacons, this makes the maximum area of the beacons equals the number of the beacons. n_b cannot be 0, otherwise the beacons would not exist. The labels of each variable are shown in **Figure 4**, and the beacon pyramid built is shown in **Figure 5**.

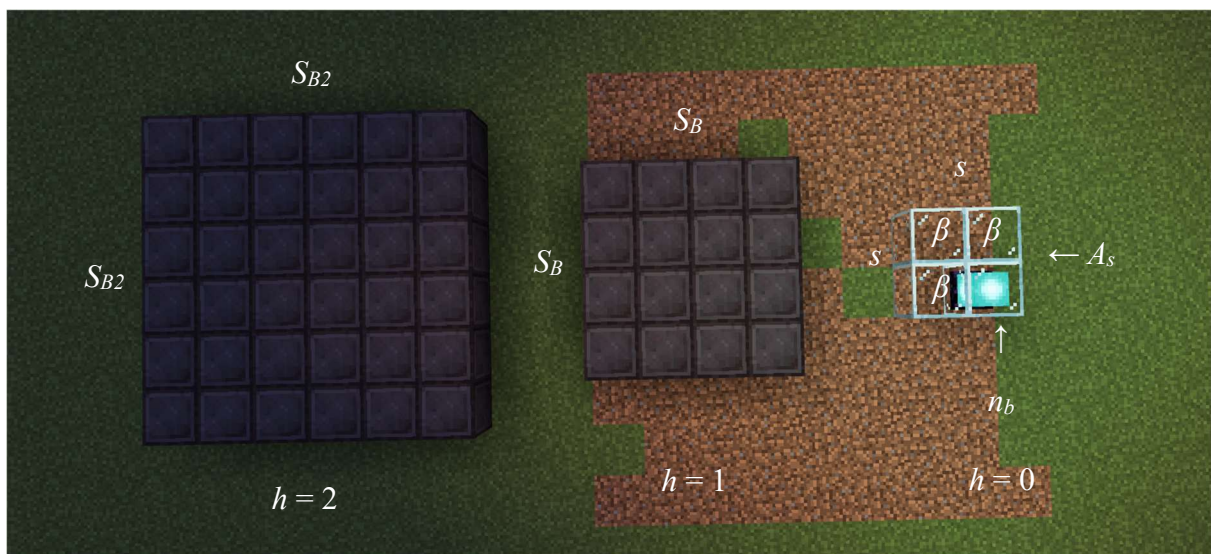


Figure 4: The beacon variables and components of $A_s = 4 \text{ blocks}^2$ from $h = 0$ and $m = 2$.

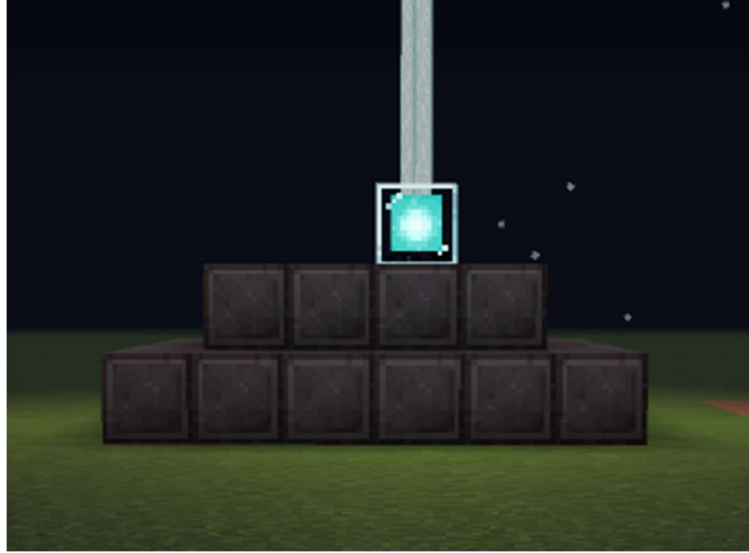


Figure 5: The beacon layers and components being fully built.

The sum of layers excluding the beacons in the second formula would be different as the number of beacons can be greater than 1. Instead of subtracting the total number of blocks to build the beacon by 1, it will be subtracted by the number of beacons.

The Sum of Layers Excluding the Beacons

$$\sigma_l = b - n_b$$

The Final Beacon Formula

The final beacon formula is the detailed formula of the beacon when the number of the maximum beacons can be any value of $\mathbb{N}$. This means that the maximum beacons in the sides can be different. Take the second formula then adjust it.

$$b_l = (2h + s)^2$$

$$b_l = (2h + s)(2h + s)$$

Since the sides are not equal, they are going to be labelled.

$$b_l = (2h + s_1)(2h + s_2)$$

Since the sides are not equal, this means that one of them is the length and the other is the width depending on the longest and the shortest side.

Let $s_1 = L$, and $s_2 = W$ to help differentiate between the length and the width.

The Final Beacon Layer Formula

$$b_l = (2h + L)(2h + W)$$

$$h \in \mathbb{N}_0$$

$$L \in \mathbb{N}$$

$$W \in \mathbb{N}$$

Where L is the length of the maximum beacons and W is the width of the maximum beacons. Both cannot equal to 0, otherwise the beacons would not exist. To find the total maximum beacons or the area of them, utilize the same formula as the second formula. The variables will only change.

$$A_s = LW$$

To find the length or the width of a beacon's layer, use the formulas below.

The Beacon Layer Length Formula

$$L_B = 2h + L$$

$$h \in \mathbb{N}_0$$

$$L \in \mathbb{N}$$

The Beacon Layer Width Formula

$$W_B = 2h + W$$

$$h \in \mathbb{N}_0$$

$$W \in \mathbb{N}$$

L_B is the length of the layer while W_B is the width of the layer. Notice that b_l can be written in terms of L_B and W_B .

$$b_l = (2h + L)(2h + W)$$

$$b_l = L_B W_B$$

If $h = 0$, $L_B = L$ and $W_B = W$. This also proves that $b_l = A_s$ when h is 0.

There is a final formula that calculates the sum of the layers and the beacons needed in a beacon pyramid. The concept has never changed; however, the variables are different since the sides do not have to share the same value.

$$b = \sum_{h=0}^m b_l - \beta$$

b_l can be written in terms of the length and the width.

$$b = \sum_{h=0}^m L_B W_B - \beta$$

This is not the aim as the summation states its initiation using the variable h . Hence, the formula should be written in terms of h . This generates the full version of the formula where the beacon layer or layer 0 can share different maximum areas depending on the player's needs and/or desire.

The Final Beacon Formula

$$b = \sum_{h=0}^m (2h + L)(2h + W) - \beta$$

$$h \in \mathbb{N}_0$$

$$L \in \mathbb{N}$$

$$W \in \mathbb{N}$$

$$m \in \mathbb{N}$$

$$\beta \in \mathbb{N}_0 \mid 0 \leq \beta < A_s$$

To summarize the entire variables in the summation, h is the number of the layer and $h = 0$ is the initiation of the beacon layers which is the number of the total maximum beacons (The smallest layer in area). L is the length of the maximum beacons measured in blocks. W is the width of the maximum beacons also measured in blocks. m is the finalization of the layer which is the biggest layer in area. Finally, β is the number of the deducted beacons. This makes b the total blocks needed in the beacon pyramid including the beacons calculating the required blocks precisely.

To find the deducted beacons, utilize the equation of the deducted beacons. The equation is unchanged, but it can be represented in terms of L_B and W_B .

The Equation of the Deducted Beacons

$$\beta = A_s - n_b$$

$$n_b \in \mathbb{N} \mid n_b \leq A_s$$

Remember that n_b is the number of the beacons desired by the player. If $\beta = 0$, $A_s = n_b$ meaning that the player chose the number of beacons to equal the total maximum area of the beacons; however, $A_s \neq \beta$ otherwise the beacons would not exist. Likewise, $n_b \neq 0$. **Figure 6** shows the entire variables and components of the beacon pyramid.

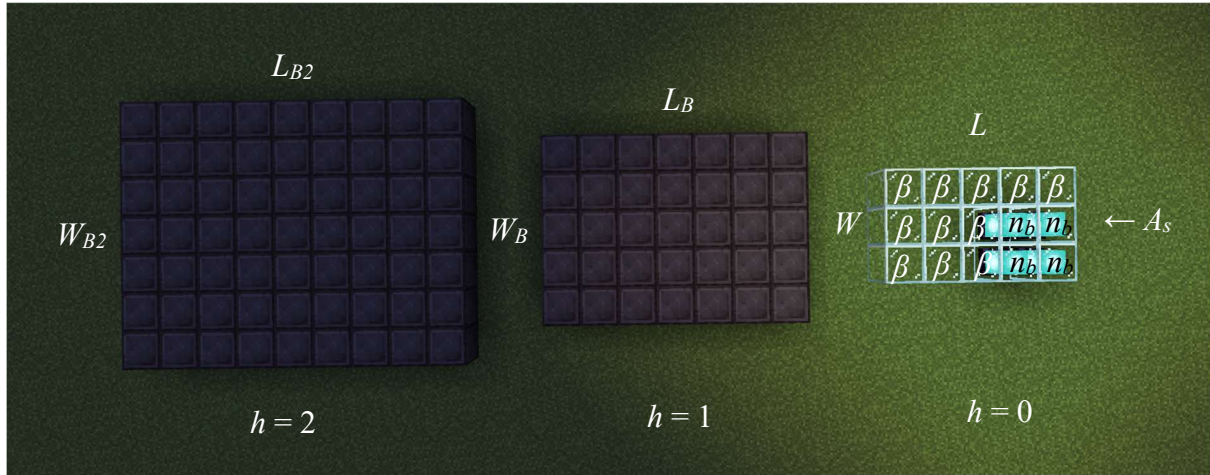


Figure 6: Variables and components of the beacon pyramid with $A_s = 15 \text{ blocks}^2$ from $h = 0$ to $h = 2$.

Figure 7 proves the validity of the components showing the desired beacons activated in a side view.



Figure 7: The beacon pyramid being fully built in a side view.

This side is not enough to prove the validity of the components.
Hence, **Figure 8** shows the same pyramid in a top view.

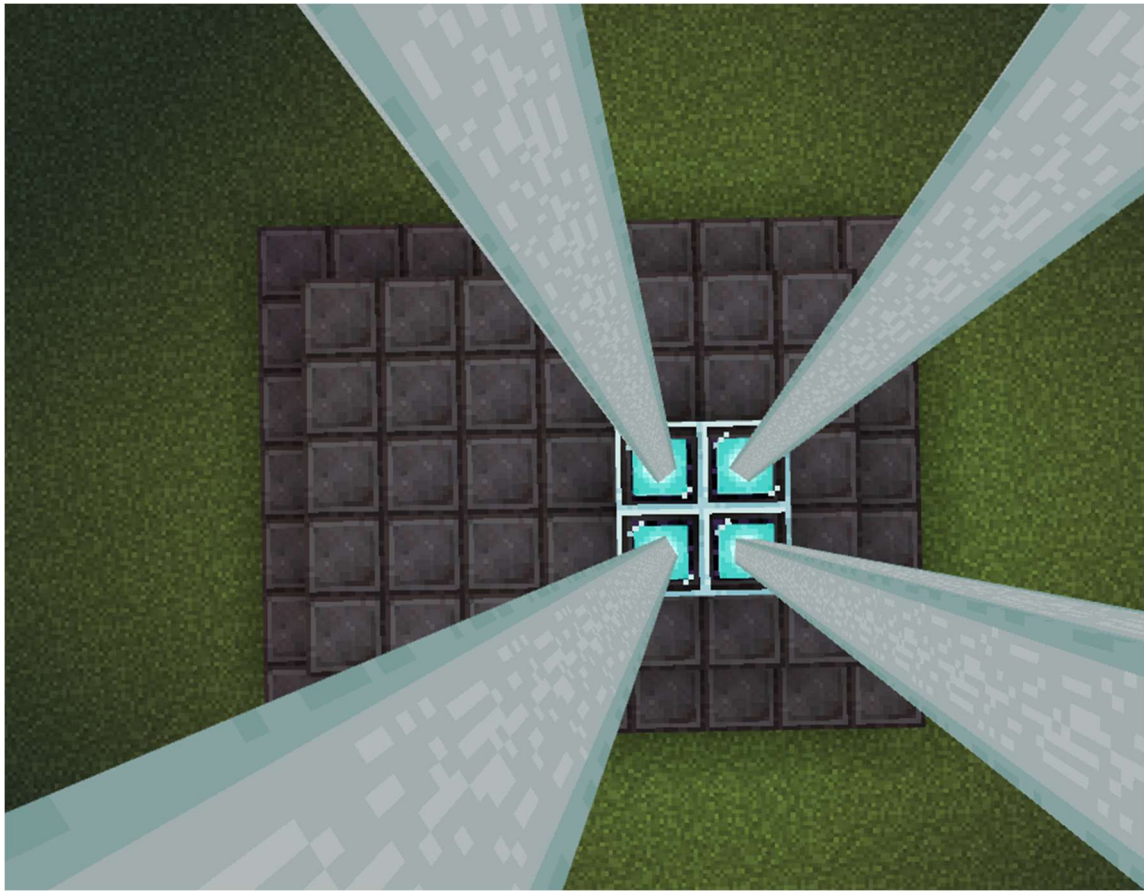


Figure 8: The beacon pyramid in a top view.

To finalize the theorem, a final formula will be mentioned. It is not a new formula; however, it accurately calculates the number of blocks needed to build the layers excluding the number of beacons. The variables of this formula are unchanged.

The Sum of Layers Excluding the Beacons

$$\sigma_l = b - n_b$$

Summary

The beacon is a powerful block used to provide effects to the player obtained by killing the wither boss. There are specific minerals used to build the beacon and activate its effects. The more the layer increased, the more the choices of the effects increase and the higher the range of effects becomes.

There are 3 laws of formulas that calculate the blocks of the beacon's layer, and the total blocks of the beacon pyramid including or excluding the beacons.

If the total maximum beacons are greater than 1, the number of beacons varies from the total maximum beacons generating the deducted beacons which is the difference between them.

Optional Questions

1. A 5-layer square beacon pyramid is built having 2 beacons placed out of a maximum of 16 blocks². Determine the blocks needed to build the beacon including and excluding the beacons.
2. A beacon has its 3rd layer to be 91 blocks². Determine the length and the width of the maximum beacons and find the area of the total maximum beacons.
3. If $A_s = 5$ blocks², find the total blocks needed to build a 7-layer beacon pyramid including the beacons. Assume that $\beta = 0$.
4. If a beacon pyramid with $A_s = 1$ blocks², determine the total blocks needed in layer 21, and calculate the total blocks needed to build a beacon pyramid with a height of 22 blocks including and excluding the beacon(s).